Experiment Name :8-bit Parallel Subtractor Circuit Design and Observation

1. Objective

To design and simulate an 8-bit Parallel Subtractor Circuit using IC 74LS83 and XOR Gates with the 2's Complement method.

2. Apparatus / Tools

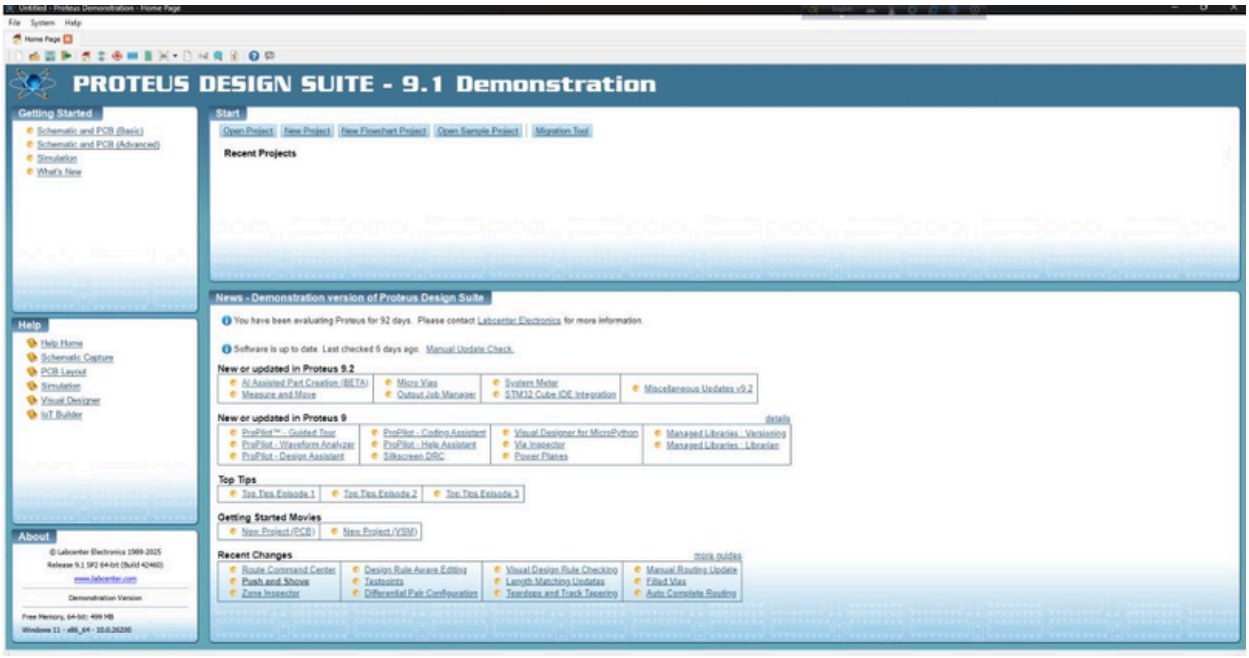
- Proteus Simulation Software
- IC 74LS83 – 2 Pieces
- IC 74LS86 (XOR Gate) – 2 Pieces
- Logic Toggles & LEDs

3. Theory:

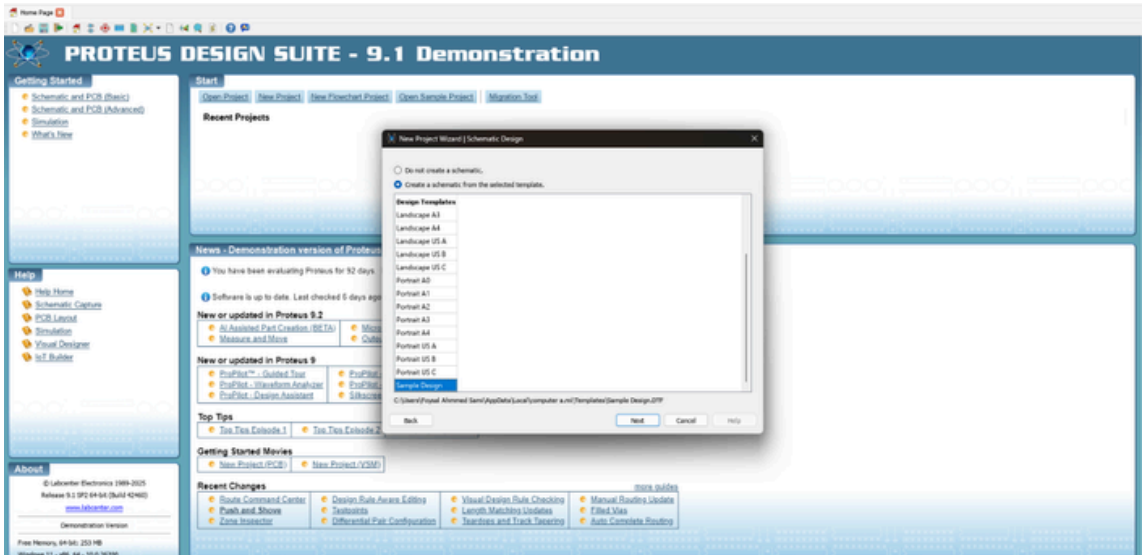
Subtraction (A − B) is performed by calculating A + (2's Complement of B). The B input is first complemented using XOR gates, and by applying 1 to the carry input (C0) of the first IC, the 2's Complement operation is completed.

4. Circuit Connection / Working Procedure

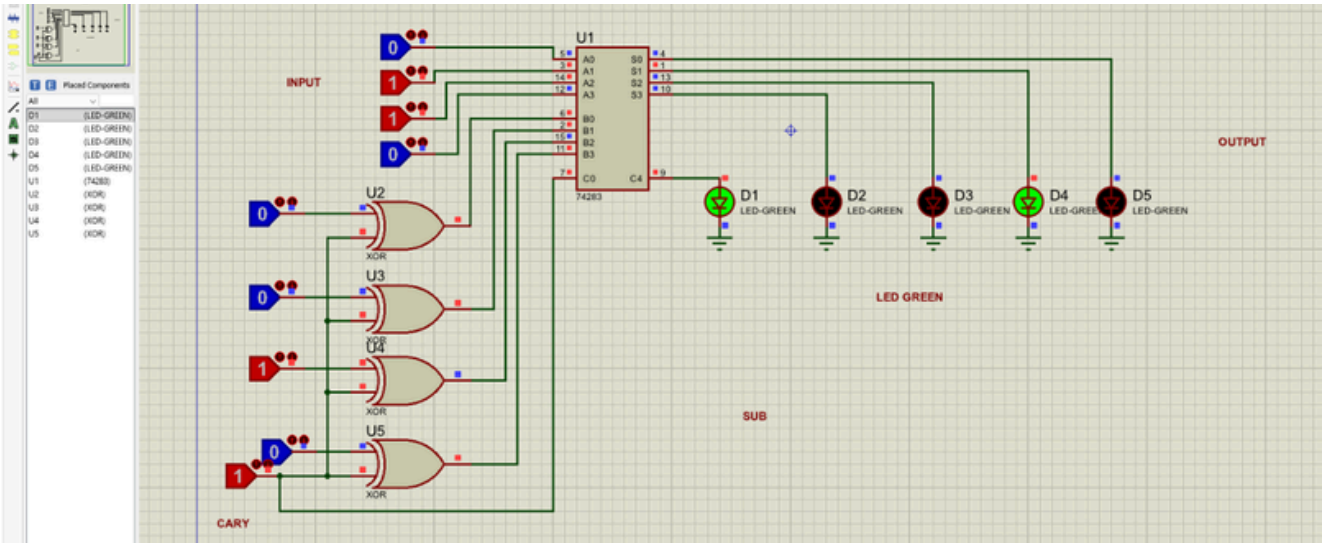
Step 1: Open the Proteus Design Suite software and click on the New Project button from the Start section on the Home Page.



Step 2 : Select the "Create a schematic from the selected template" option, choose Sample Design (or DEFAULT) from the Design Templates list, and click Next to open the workspace.



Step 3 : Pick the required components (74283, 74LS86, LOGICSTATE, LED-GREEN, GROUND) using Component Mode (P), connect XOR gates to B inputs with control line M=1, complete the cascaded wiring, and press the Play button to run the simulation.



Conclusion: The 8-bit Parallel Subtractor circuit was successfully simulated using the 2's complement method, giving correct subtraction results on the LEDs.

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